

CSCI 5322

Algorithmic Human-Robot Interaction

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	AI Engineer (72.3 / 100)	55.9 / 100

56	91	+47
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Creating autonomous systems that interact with humans requires the synthesis of insights from a variety of disciplines. This course aims to provide students with the algorithms, models, and frameworks that form the building blocks required for developing intelligent autonomous systems that perform useful tasks while interacting with, coordinating with, co-existing with, or otherwise assisting humans. Previously offered as a special topics course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	72.3	100.0	+27.7	MEDIUM
Machine Learning Engineer	61.8	100.0	+38.2	HIGH
ML Engineer	61.5	88.5	+27.0	MEDIUM
Applied Scientist	60.3	100.0	+39.7	HIGH
Data Scientist	41.7	88.4	+46.6	HIGH
MLOps Engineer	37.9	68.6	+30.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Model Deployment & Serving 2 of 6 roles

production

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **MEDIUM** Consider Model Deployment & Serving content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

GAP KEYWORDS BY ROLE

AI Engineer

72.3 → 100.0 (+27.7 pts)

data robotics engineering learning build software technical problems

Machine Learning Engineer

61.8 → 100.0 (+38.2 pts)

data engineering learning tools solutions time design science

ML Engineer

61.5 → 88.5 (+27.0 pts)

media data learning machine production engineering starting build

Applied Scientist

60.3 → 100.0 (+39.7 pts)

lab research problems technical data learning engineering design

Data Scientist

41.7 → 88.4 (+46.6 pts)

data product characteristics evolution different social build engineering

MLOps Engineer

37.9 → 68.6 (+30.7 pts)

data production text learning machine media engineering tools

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.